Question ID: 4618501a

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Nonlinear functions	Medium

Question

$$f(x) = 3{,}000(0.75)^x$$

A conservation scientist implemented a program to reduce the population of a certain species in an area. The given function estimates this species' population x years after 2008, where $x \leq 8$. Which of the following is the best interpretation of $3{,}000$ in this context?

Answer

A. The estimated percent decrease in the population for this species and area every 8 years after 2008

B. The estimated percent decrease in the population for this species and area each year after 2008

C. The estimated population for this species and area 8 years after 2008

D. The estimated initial population for this species and area in 2008

Correct Answer: D

Rationale

Choice D is correct. Substituting 0 for x in the given equation yields $f(0) = 3{,}000(0.75)^0$, which is equivalent to $f(0) = 3{,}000(1)$, or $f(0) = 3{,}000$. It's given that the function estimates the species' population x years after 2008, so it follows that the estimated population of the species is $3{,}000$ in 2008. Therefore, the best interpretation of $3{,}000$ in this context is the estimated initial population for this species and area in 2008.

Choice A is incorrect and may result from conceptual errors.

Choice B is incorrect. The estimated percent decrease in the population for this species and area each year after 2008 is 25%, not $3{,}000$.

Choice C is incorrect. The estimated population for this species and area 8 years after 2008 is $3{,}000(0.75)^8$, or approximately 300, not $3{,}000$.